

Suhan Ramani

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EDUCATION

Indian Institute of Technology, Jodhpur

B.Tech in Computer Science and Engineering | CGPA: 7.85/10

Coursework: Data Structures & Algorithms, Software Design, Pattern Recognition & Machine Learning, Probability & Statistics, Mathematics for Computing

Aug 2024 – Jul 2028

Jodhpur, Rajasthan

Ashadeep Vidyalay

Class XII (GSEB): 94.00% | Class X (GSEB): 92.50%

2022 – 2024

TECHNICAL SKILLS

Languages: C/C++, Python, JavaScript, TypeScript, Java, PHP, Dart, SQL

Frameworks & Libraries: FastAPI, React.js, Flutter, Node.js, TensorFlow, MediaPipe, Riverpod

Systems & Infrastructure: Distributed systems, cloud-native application development, multithreading, concurrency & synchronization, async I/O, event-sourced architectures, PostgreSQL, MySQL, MongoDB, Qdrant (Vector DB), Cloud Firestore, Firebase

AI/LLM Engineering: Retrieval-Augmented Generation (RAG), Prompt Engineering, LLM Context Management, Gemini API, AI-Assisted Development (Claude)

Tools & Domains: Git, Linux, REST APIs, Agile/Scrum | Software Design, System Architecture, Machine Learning, AI-integrated systems, performance analysis & debugging, data structures & algorithms

WORK EXPERIENCE

Software Development Engineer Intern

May 2026 – Jun 2026

Webify Design | Web Development & E-Commerce Agency

PHP, MySQL, JavaScript, HTML5/CSS3, Shopify

- Shipped end-to-end features across 5+ **production web applications** using PHP, HTML5/CSS3, and JavaScript for desktop and mobile, using AI-assisted development (Claude) to speed up iteration; reused existing modules across similar client requirements (e.g., cart/checkout flows) to cut delivery time on repeat features instead of rebuilding from scratch.
- Optimised scalable MySQL schemas with complex queries and stored procedures, restructuring indexes and eliminating N+1 patterns across high-traffic endpoints to **reduce data-retrieval latency by ~35%**, directly improving page responsiveness for end users.
- Collaborated across teams to debug and resolve critical production incidents via root-cause analysis on regressions and DB bottlenecks, improving infrastructure reliability across deployed systems.

Student Developer

Dec 2025 – Apr 2026

SQORA | AI-Powered Learning Platform (Winter of Code)

Python, FastAPI, Qdrant, Gemini API

- Architected and deployed a production-grade RAG pipeline with prompt engineering and LLM context management, integrating Gemini API and Qdrant Vector DB – indexed **500+ documents** with end-to-end latency **under 500 ms**, serving 3 cross-functional engineering workflows with strong retrieval relevance.
- Constructed concurrent RESTful APIs in FastAPI with async/await non-blocking I/O, eliminating concurrency bottlenecks across **4 parallel ingestion streams**; built validation pipelines that eliminated all observed embedding failures during testing.

RESEARCH AND PROJECTS

Communication-Efficient Leader Election via Hybrid Probabilistic Agreement | In-Progress Paper Distributed Systems

- Co-authored a hybrid distributed systems framework under Prof. Nitin Awathare (IIT Jodhpur CSE) for leader election in a Byzantine-faulty system (targeting Dec 2026 submission), using cryptographic digital signatures and a public-key infrastructure to achieve **Byzantine fault-tolerant consensus** ($f < N/3$ static faults).
- Proved that committee membership can be derived locally by every node from a shared pseudorandom function and the public round number, requiring zero communication for selection while remaining **statistically indistinguishable** from a uniformly random committee, even against a static Byzantine adversary.

Real-Time Sign Language Detection & Sequence Recognition | GitHub Python, TensorFlow, FastAPI, React.js, MediaPipe

- Developed a sequence-aware Transformer model applying machine learning and computer vision to achieve **96%+ accuracy** across 26 ASL gesture classes trained on 30,000+ landmark frames; normalised features to decouple accuracy from lighting variance for robust real-world use.
- Deployed a low-latency FastAPI inference microservice streamed to React.js via WebSocket at sub-100 ms per frame; boosted throughput **40%** via batch-processing and model quantisation, cutting model size **4x** with **<2%** accuracy loss.

Spendly – Expense Ledger & Debt Simplification | GitHub

Flutter, Dart, Riverpod, Cloud Firestore

- Designed an event-sourced, append-only ledger across 5+ concurrent users per session, eliminating concurrency race conditions by construction (no locking) and achieving **synchronization guarantees equivalent to CRDT patterns** with no observed data conflicts in testing.
- Devised a Minimize Cash Flow – $O(n \log n)$ algorithm reducing group settlements to at most $n-1$ transactions for n members (**up to 70% fewer** payments vs. naive approach); orchestrated 16 concurrent Firestore listeners with Riverpod across 20+ screens.
- Collaborated cross-functionally within the **Alpha Team** (with Krish Patel, Maitry Gami, and Diya Limbani) in an Agile/Scrum workflow with sprint-based iterations, contributing to the Software Requirements Specification (SRS) to align user requirements ahead of implementation.

POSITIONS OF RESPONSIBILITY

Overall Coordinator, Anand Rathi Tinkerers' Lab (ARTL), IIT Jodhpur

Aug 2026 – Present

Lead the student body of IIT Jodhpur's 3,000+ sq. ft. student-run makerspace, coordinating campus access and project support for the campus maker community.

Core Team Member, Programming Society (P-Club), IIT Jodhpur

Aug 2025 – Present

Crafted **10+ algorithmic contest problems** using Codeforces Polygon, including test data generation and validation; mentored juniors on competitive programming and scalable optimisation strategies.

Core Team Member, Office of Training & Placement, IIT Jodhpur

Oct 2025 – Present

Manage corporate outreach to tier-1 firms; collaborate cross-functionally to coordinate large-scale placement pipelines.

Core Team Member, Entrepreneurship Cell (E-Cell), IIT Jodhpur

Sep 2024 – Present

Organised **5+ campus-wide events** driving participation of **300+ students**; mentored a student team for **Vandre**, the institute's annual entrepreneurship fest.

ACHIEVEMENTS AND LEADERSHIP

- Competitive Programming:** Solved **500+** problems across LeetCode, GeeksforGeeks & Codeforces (Max rating:**1319**), demonstrating strong algorithms & software design fundamentals.
- JEE Advanced 2024:** Secured All India Rank (AIR) **397** in the General-EWS category.
- JEE Main 2024:** Achieved **99.55 percentile** nationally out of approximately **1.2 million** candidates.
- GUJCET 2024:** State Rank **3** in Gujarat.
- Vandre Entrepreneurship Fest, IIT Jodhpur:** Won **Best Pitch** and **2nd Runner-Up** for a self-made product stall over a 2-day pitch-and-sell event.
- FLY-Scholar (Finding the Leader in You) Program:** Certified by the Competitiveness Mindset Institute, USA, for developing leadership and non-cognitive skills – conscientiousness, initiative, perseverance, and problem-solving.
- Swami Dayanand Education Foundation Scholarship:** Recipient of a national merit-cum-means scholarship for academic excellence.
- Extra-Curricular:** Badminton and Cycling.